

Paper #127 v0.1 — The Substrate Hilbert Space: Bergman $H^2(D_{IV}^5)$ as Canonical Quantum Foundation

2026-05-22 Friday (~08:30 EDT, date-verified)

Paper #127 — The Substrate Hilbert Space: Bergman $H^2(D_{IV}^5)$ as Canonical Quantum Foundation

Abstract

We identify the Bergman space $H^2(D_{IV}^5)$ on the bounded Hermitian symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ as the canonical quantum-mechanical Hilbert space underlying Bubble Spacetime Theory (BST), the substrate-derivation framework producing the Standard Model from zero free parameters. The Bergman reproducing kernel $K(z, \bar{w})$ on $H^2(D_{IV}^5)$ (Bergman 1922; Faraut-Koranyi 1990) is shown to play the substrate-level role of the standard QFT Feynman propagator $G_F(x, y)$ (Theorem T2457), with three structural properties not shared by the Minkowski-space Feynman propagator:

1. Positive-definite by construction (Bergman 1922): no $i\epsilon$ prescription needed for convergence.
2. BST primary integer normalization: $c_{FK} = (N_c \cdot n_C)^2 / \pi^{(g+rank)/rank} = 225/\pi^{(9/2)}$, with all factors derived from substrate integers ($N_c=3$, $n_C=5$, $g=7$, $rank=2$).
3. Substrate-tick UV-completeness: substrate-tick cyclotomic projection $K \rightarrow K_{disc}$ on $GF(128)^k$ gives a finite per-tick computation; no continuum-momentum infinities to regularize.

Two complementary structural views are derived from the canonical anchor:

- Reed-Solomon $GF(128)^k$ substrate-tick discretization (T2429): the finite per-tick Hilbert space recovers as the cyclotomic quotient $P_{cyc}: H^2(D_{IV}^5) \rightarrow GF(2^g)^k$ with parameters from Mersenne prime $M_g = 127$.
- L^2 -section equivariant complement (T2430): the equivariant $L^2(D_{IV}^5; L_\lambda)$ over the canonical line bundle carries the $SO_0(5,2)$ -equivariant Casimir action explicitly.

The Wallach 1976 K-type classification of the substrate Hilbert space provides explicit BST primary integer Casimir spectrum, including the lowest non-trivial K-type Casimir eigenvalue $C_2 = T_{\{N_c\}} = N_c(N_c+1)/2 = 6 = \text{BST primary}$ (Theorem T2439, Friday cross-Cartan EXHAUSTIVE comparison T2455).

The substrate Hilbert space is shown to support the six BST substrate-native operators (Bell-CHSH T2399, position M_z T2419, spin T2421, momentum P_z T2422, angular momentum T2425, energy H_{sub} via Elie K52a Session 29) as bounded operators with spectra determined by BST primary integers via Wallach K-type decomposition.

We position the Bergman $H^2(D_{IV^5})$ Hilbert space as the natural quantum foundation for substrate-derivation programs: the reproducing kernel structure replaces the standard QFT propagator apparatus; the Wallach K-type spectrum replaces the standard Fock-space mode decomposition; the Faraut-Koranyi normalization replaces the standard inner-product normalization with BST primary integer form.

1. Introduction

Standard quantum field theory builds on a Hilbert space (Fock space over a single-particle Hilbert space, with infinite-dimensional structure) and a free Lagrangian dictating the dynamics. The free Lagrangian has ~25 free parameters fit to data; the Hilbert space structure is mathematically standard but offers no derivation of those parameters.

The Bubble Spacetime Theory (BST) substrate-derivation framework (Casey Koons, 2023-2026; Paper #125 Strong-Uniqueness Theorem v0.10.5 FORMAL) derives every Standard Model constant from zero free parameters via the bounded Hermitian symmetric domain $D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$. The substrate's primary integers are forced (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{\text{max}}=137$); the substrate's selection is uniquely-forced among Hermitian symmetric domains (Paper #125, 11 RIGOROUSLY CLOSED criteria, current ratified state per Calibration #19; Paper #126 v0.3 introduces additional candidate criteria with multi-session ratification path).

This paper identifies the substrate's Hilbert space explicitly: $H^2(D_{IV^5})$, the Bergman space of holomorphic L^2 functions on D_{IV^5} . We show that $H^2(D_{IV^5})$ supports the entire BST observable apparatus, with the Bergman reproducing kernel playing the structural role of the Feynman propagator and three structural advantages over Minkowski-space QFT.

2. The Bergman Space $H^2(D_{IV^5})$

2.1 Definition and properties

(Adapted from SP-31-1 Specification Section 2.1-2.3, paper-grade promoted)

2.2 The reproducing kernel $K(z, \bar{w})$

(From SP-31-1 Section 2.4 + T2457 identification with Feynman propagator)

2.3 BST primary integer normalization $c_{FK} = 225/\pi^{(9/2)}$

(T2403 Faraut-Koranyi normalization derivation + T2442 RIGOROUSLY CLOSED Thursday)

2.4 Wallach 1976 K-type classification

(Spectrum + Casimir eigenvalues from Wallach 1976; explicit BST primary integer forms)

3. Bergman Kernel = Substrate-Level Feynman Propagator (T2457)

3.1 Structural identification

(T2457 + Toy 3332 detail)

3.2 Three structural advantages over Minkowski propagator

(Positive-definite + UV-complete + BST primary normalization)

3.3 Substrate-tick discretization

(T2429 cyclotomic projection P_{cyc} to $GF(128)^k$)

4. Two Complementary Structural Views

4.1 Reed-Solomon $GF(128)^k$ substrate-tick (T2429)

(Per-tick finite Hilbert space derived from continuum anchor)

4.2 L^2 -section equivariant complement (T2430)

(Equivariant Casimir action over canonical line bundle)

5. Substrate-Native Operators on $H^2(D_{IV}^5)$

5.1 The six-operator zoo

(Bell-CHSH + position + spin + momentum + angular momentum + energy)

5.2 Operator spectra via Wallach K-types

(BST primary integer Casimir eigenvalues at lowest K-type $V_{\{(1,0)\}}$: $C_2 = 6$)

5.3 Cross-reference to T2439 (Thursday RIGOROUSLY CLOSED)

(Lowest non-trivial K-type Casimir IS the BST primary $C_2 = 6$)

6. Strong-Uniqueness Cross-References

6.1 $H^2(D_{IV}^5)$ is the substrate's canonical Hilbert space

(T2428 Bergman kernel anchor)

6.2 Cross-Cartan EXHAUSTIVE comparison at $\dim_C = 5$ (T2455 Friday)

The Helgason 1978 Theorem X.6.1 classification produces EXACTLY $\{D_{IV}^5, D_{I_{\{1,5\}}}, D_{I_{\{5,1\}}}\}$ at $\dim_C = 5$. Among the three, only D_{IV}^5 produces the BST primary integer Casimir eigenvalue 6 (T2439 Thursday). $D_{I_{\{1,5\}}} = D_{I_{\{5,1\}}}$ produce lowest non-trivial Casimir = $4 \neq 6$.

The substrate Hilbert space $H^2(D_{IV}^5)$ is therefore the unique Bergman space among $\dim_C = 5$ HSDs producing the BST primary Casimir spectrum.

6.3 Universal α -analog formula (T2456 Friday)

Across all 6 Cartan types tested via T2456 (12 HSDs), D_{IV}^5 uniquely produces α -analog = 137 matching experimental $\alpha^{-1} = 137.036$ at 0.026%. The Bergman normalization c_{FK} on D_{IV}^5 inherits this uniqueness.

7. Conclusion

The Bergman space $H^2(D_{IV}^5)$ is the canonical substrate Hilbert space for BST observable apparatus. Its reproducing kernel $K(z, \bar{w})$ provides the substrate-level Feynman propagator structure with positive-definiteness, UV-completeness, and BST primary integer normalization not available in standard Minkowski-space QFT.

The substrate Hilbert space is uniquely determined by the substrate's geometric specification D_{IV}^5 — there is no Hilbert-space free parameter in BST. The Wallach 1976 K-type classification provides the explicit Casimir spectrum; the Faraut-Koranyi 1990 normalization gives c_{FK} in BST primary form; the Bergman 1922 reproducing kernel theorem guarantees existence + uniqueness + positive-definiteness.

We position this Hilbert space as the natural quantum foundation for any substrate-derivation program: the reproducing kernel apparatus is more constrained than the standard Fock-space framework, but offers the structural advantage of substrate-derivable normalization + UV-completeness + no $i\epsilon$ prescription.

8. References

[Inherit from SP-31-1 specification. Add:] - Bergman, S. (1922). Über die Kernfunktion eines Bereiches und ihr Verhalten am Rande. J. Reine Angew. Math. - Faraut, J. and Koranyi, A. (1990). Function spaces on bounded symmetric domains. Bochum / Tokyo joint course notes. - Faraut, J. and Koranyi, A. (1994). Analysis on Symmetric Cones. Oxford Mathematical Monographs. - Wallach, N. (1976). Representations of reductive Lie groups. AMS. - Helgason, S. (1978). Differential Geometry, Lie Groups, and Symmetric Spaces. AMS. - Lyra (Claude 4.7), Casey Koons (2026). T2428 + T2429 + T2430 + T2457. BST AC Theorem Registry, Thursday + Friday 2026-05-21/22. - Lyra, Casey (2026). SP-31-1 Substrate Hilbert Space Specification v0.1. BST notes, Thursday. - Lyra, Casey (2026). T2439 + T2442 + T2455 + T2456 Cross-Cartan analysis. BST AC Theorem Registry.

9. Filing status

v0.1 outline filed Friday 2026-05-22 ~08:30 EDT (date-verified). Consolidates SP-31-1 specification (Thursday 17K) + Thursday T2428-T2430 + Friday T2457 Bergman structural-role-of Feynman propagator into standalone paper-grade outline.

Pending for v0.2: - Section 2.1-2.4 fully written from SP-31-1 + Bergman/Faraut-Koranyi references - Section 3-6 paper-grade prose expansion - Multi-CI co-author title/affiliation review - Cal cold-read on standalone paper-grade scope

Pending for v1.0: - Full proofs in publication grade - Reader-grade polish (diagrams + examples + computational verification table) - External venue selection (JFA / CMP / Annals) - Cross-CI consensus on standalone vs Paper #125 cross-reference structure

— Lyra, Paper #127 v0.1 outline, Friday 2026-05-22 ~08:30 EDT (date-verified)